

Power Series

Complex Analysis

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Definition

Let $\Omega \subseteq \mathbb{C}$ and suppose that for each $n \in \mathbb{N}$ there is a function $f_n : \Omega \rightarrow \mathbb{C}$; we shall say that (f_n) is a **sequence of functions** on Ω to $\mathbb{C}$.

Definition

Let (f_n) be a sequence of functions on $\Omega \subseteq \mathbb{C}$ to $\mathbb{C}$, let $\Omega_0 \subseteq \Omega$, and let $f : \Omega_0 \rightarrow \mathbb{C}$. We say that f the sequence (f_n) **converges on Ω_0 to f** if, for each $z \in \Omega_0$, the sequence $(f_n(z))$ converges to $f(z)$ in $\mathbb{C}$. In this case we call f the **limit on Ω_0** of the sequence (f_n) . When such a function exists, we say that the sequence (f_n) **converges pointwise on Ω_0** .

Definition

If (f_n) is a sequence of functions defined on a subset Ω of $\mathbb{C}$ with values in $\mathbb{C}$, the sequence of **partial sums** (s_n) of the infinite series $\sum f_n$ is defined for z in Ω by

$$s_1(z) := f_1(z)$$

$$s_2(z) := s_1(z) + f_2(z)$$

$$\vdots$$

$$s_{n+1}(z) := s_n(z) + f_n(z)$$

$$\vdots$$

In case the sequence (s_n) of functions converges on Ω to a function f , we say that the infinite series of functions $\sum f_n$ **converges** to f on Ω . We will often write

$$\sum f_n \quad \text{or} \quad \sum_{n=1}^{\infty} f_n$$

to denote either the series or the limit function, when it exists.

Definition

If the series $\sum |f_n(z)|$ converges for each $z \in \Omega$, we say that $\sum f_n$ is **absolutely convergent** on Ω . If the sequence (s_n) of partial sums is uniformly convergent on Ω to f , we say that $\sum f_n$ is **uniformly convergent** on Ω , or that it **converges to f uniformly on Ω** .

Theorem 1.1

Let (f_n) be a sequence of complex functions on Ω . If the series of functions $\sum f_n$ is absolutely convergent, then it is convergent.

Theorem 1.2 (Weierstrass Criterion)

Let (f_n) be a sequence of complex functions on Ω . If for every integer n , there exists $M_n > 0$ with the properties $|f_n(z)| \leq M_n$ for all $z \in \Omega$ and $\sum M_n$ is convergent, then the series of functions $\sum f_n$ is uniformly convergent.

Definition

A series of complex functions $\sum f_n$ is said to be a **power series around** $z = z_0$ if the function f_n has the form

$$f_n = a_n(z - z_0)^n,$$

where a_n and z_0 belong to $\mathbb{C}$ and where $n = 0, 1, 2, \dots$

Definition

Let (a_n) be a sequence in $\mathbb{C}$. If the sequence $(|a_n|^{1/n})$ is bounded above, we set $\rho := \limsup |a_n|^{1/n}$; if this sequence is not bounded above we set $\rho = \infty$. We define the **radius of convergence** of $\sum a_n(z - z_0)^n$ to be given by

$$R := \begin{cases} 0 & \text{if } \rho = \infty, \\ 1/\rho & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

This is known as **Hadamard's formula**.

If R is the radius of convergence of a power series around z_0 , then the **disc of convergence**[†] is the set $D_R(z_0)$.

[†]This definition is due to Niels Henrik Abel.

Theorem 1.3 (Abel-Cauchy-Hadamard's Theorem)

Given the power series $\sum a_n(z - z_0)^n$ with $a_n, z, z_0 \in \mathbb{C}$, define the number $R \in [0, \infty]$ by Hadamard's formula. Then

- i) if $|z - z_0| < R$, the series converges absolutely;
- ii) if $|z - z_0| > R$, the series diverges (the terms of the series become unbounded);
- iii) if $0 < r < R$ the series converges uniformly throughout $\overline{D}_r(z_0)$.

Remark

It should be noted that the Cauchy-Hadamard Theorem makes no statement as to whether the power series converges when $|z| = R$. Indeed, anything can happen, as the examples

$$\sum x^n, \quad \sum \frac{1}{n} x^n, \quad \sum \frac{1}{n^2} x^n, \quad x \in \mathbb{R}$$

show. Since $\lim(n^{1/n}) = 1$, each of these power series has radius of convergence 1. The first power series converges at neither of the points $x = -1$ and $x = 1$; the second series converges at $x = -1$ but diverges at $x = 1$; and the third power series converges at both $x = -1$ and $x = 1$.

Theorem 1.4 (Ratio Criterion)

Let $\sum a_n(z - z_0)^n$ be a power series with radius of convergence R . Then

$$R = \lim \left| \frac{a_n}{a_{n+1}} \right|$$

provided that the limit exists or properly diverges.

Example 1.5

Consider the series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}, \quad \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}.$$

The radius of convergence of each series is infinite ($R = \infty$), that is, each series converges (absolutely and uniformly) on $\mathbb{C}$.

Definition

Let Ω be an open subset of $\mathbb{C}$ and let $z_0 \in \Omega$. A function $f : \Omega \rightarrow \mathbb{C}$ is said to be **analytic at z_0** if there exists a power series $\sum a_n(z - z_0)^n$ around z_0 with radius of convergence $r > 0$ such that $B_r(z_0) \subset \Omega$ and

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n \quad \text{for all } z \in B_r(z_0).$$

Further, we say that f is **analytic** if it is analytic at each point of Ω . Furthermore, the family of all analytic functions on Ω will be denoted by $\mathcal{A}(\Omega)$.